

Nithin Eswarappa

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Computer Vision: Expert in multimodal image fusion (Visible-Infrared-SAR) and end-to-end detection and segmentation pipelines; developed novel SOTA-beating fusion algorithms spanning bi-modal and tri-modal synthesis, cross-attention architectures, and multi-layer feature alignment, with results accepted at conferences IEEE TENCON, IEEE SPACE, and IEEE CONECCT. **Deep Learning:** Designed novel architectures with custom loss functions, training pipelines, and benchmarking frameworks; applied knowledge distillation and model compression to achieve **3× faster inference** and **2× improved image quality metrics** over SOTA 2025–2026 fusion methods, with reduced model size. **Edge Deployment:** Brings research to real-world operation by optimising and deploying trained DL models as **TensorRT engines on Nvidia Jetson**, with hands-on experience in the embedded constraints—memory, latency, and power—that govern real-time inference on resource-limited platforms.

RESEARCH EXPERIENCE

MSIF LAB, GITAM UNIVERSITY | Research Associate

Bengaluru | Oct 2023 – Mar 2026

- Conducted research and development for novel and scalable implementations of fusion models for 2 modes image fusion (Visible- IR), 3 modes image fusion (Visible, Infrared, SAR) for segmentation and object detection. Designed and evaluated using CNNs, ViT, YOLO, DINO, SegFormer, and other Vision Foundation Models (VFM).
- Developed a novel scalable DL framework with knowledge distillation, achieving 3× faster inference, 2× better image quality metrics, and least model size in comparison to SOTA fusion methods.
- Prepared a benchmark Vis-IR-SAR trimodal dataset from satellite images and developed a framework for trimodal fusion and evaluation.
- Deployed optimized TensorRT inference engines on Jetson for real-time edge performance.
- Supervising:** Supervised a junior research fellow to build and deploy GUI for 2 modes image fusion with fusion performance metrics, and a pairwise comparison-based subjective image quality evaluation interface.
- Mentoring:** Guided 10 students (One Master's, nine undergraduates) on projects spanning multimodal fusion, image registration, object detection, stereo disparity estimation, and code optimization for deployment on Jetson-based embedded platforms.

PROFESSIONAL EXPERIENCE

VIMAAN.AI | Senior Embedded Engineer - Robotics

Bengaluru | Apr 2020 – Jul 2022

- Developed 6-DOF absolute localization for indoor drone navigation using ArUco marker-based camera pose estimation. Built relative pose estimation pipeline using stereo Intel RealSense depth cameras, which decreased localization drift improving warehouse coverage.
- Achieved continuous autonomous drone operations with development of automated drone battery recharging via a ArUco tag based landing and BMS.

TONBO IMAGING | Member of Technical Staff

Bengaluru | Apr 2015 – Jun 2018

- Developed optical/mechanical motion blur correction mechanism by stabilizing the line of sight for narrow-FOV dual electro-optic (Thermal IR, visible) sensors on a continuously rotating multi-sensor imaging platform. Achieved 360°/sec.
- Optimized dual stream 360°panoramic stitching pipeline from line-of-sight stabilized images and achieved 1 FPS under embedded platform constraints. Designed and developed a synchronization mechanism, for camera Vsync and the fast steering mirror for motion blur correction.
- Developed motion blur correction algorithms using point spread functions for rotating multimodal imaging platforms. Developed image processing algorithms: rolling shutter correction, non-uniformity correction, and automatic gain control for thermal imagery.
- Developed Geo-pointing capability for Pan Tilt Unit (PTU) imaging systems, integrating GPS, digital magnetic compass, and ground-based RADAR.

PROFESSIONAL MEMBERSHIP

Member, IEEE (Institute of Electrical & Electronics Engineers)

Since 2024

PUBLICATIONS

- **Eswarappa N**, Jeevan KM, Waldekar S, Vivek BK, George K. "Reformulating Multimodal Fusion as Encoder-Free Multiscale Image Synthesis with VFM Guided Representation Alignment." *Planned Submission*, 2026.
- **Eswarappa N**, Jeevan KM, Waldekar S, Vivek BK, George K. "TriFIS: Trimodal Fused Image Synthesis and A Visible-Infrared-SAR Benchmark Dataset." *Planned Submission*, 2026.
- **Eswarappa N**, Jeevan KM, Waldekar S, Vivek BK, George K. "HyLapDFN: A Hybrid Approach for Infrared-Visible Image Fusion using Laplacian Pyramid and Decoder Fusion Network." *IEEE TENCON 2025* [Oral].
- **Eswarappa N**, Waldekar S, Jeevan KM, Vivek BK, George K. "Conventional vs Deep-Learning Methods for Visible and Infrared Image Fusion: A Comparison." *IEEE SPACE 2025* [Oral].
- **Eswarappa N**, Jeevan KM, Waldekar S, Vivek BK, George K. "Trimodal Image Fusion Framework for a Visible-Infrared-SAR Benchmark Dataset." *IEEE SPACE 2025* [Oral].
- **Eswarappa N**, Waldekar S, Jeevan KM, Vivek BK, George K. "A Colour-Focused Visible and Infrared Image Fusion Framework for Aiding Human Perception." *IEEE CONECCT 2024* [Oral].

EDUCATION

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, Bengaluru
Master of Technology in Intelligent Systems

December 2015

PESIT — CIS LAB | Graduate Student Researcher

Bengaluru, India | Oct 2013 – Apr 2015

Master's Thesis: *Appearance-based terrain learning and classification for autonomous mobile robot navigation*, advised by Prof. Prabhakar Mishra. Evaluated 15 visual feature representations (5 color spaces, 5 texture descriptors, and combinations) across GMM and MLP classifiers. Implemented a scalable MLP framework from scratch (configurable layers, neurons, and a generalized backpropagation routine) to systematically search 3 architectures per feature; trained via stochastic gradient descent with momentum and layer-wise learning rate scheduling, using validation-based early stopping and cross-dataset generalization testing. A key contribution of the work is the development of a calibrated pixel-to-floor homography projection module that transforms 2D semantic classifications into a 3D rover-centric metric map, effectively converting high-level visual data into actionable spatial information for physical path planning. This work provides a scalable perception pipeline that bridges the gap between raw computer vision and reliable robotic mobility in real-world settings (Grade: 190/200).

GATE Fellowship, Electronics & Communication Engineering
IIT Madras, MHRD, Government of India

March 2011

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, Bengaluru
Bachelor of Engineering in Electronics and Communication

June 2008

TECHNICAL SKILLS

- **Deep Learning Frameworks:** PyTorch, TensorFlow, ONNX, TensorRT, Knowledge Distillation
- **Computer Vision:** Multimodal / Sensor Fusion (Visible, IR, SAR), Image Synthesis, Object Detection (YOLO, DINO), Semantic Segmentation (SegFormer), Vision Transformers (ViT), Vision Foundation Models (VFMs), Image Registration, Stereo Vision, Panoramic Stitching, Image Quality Assessment
- **Programming:** Python, C/C++, CUDA (basic), MATLAB, Bash, Git, $\text{\LaTeX}$
- **Libraries / Middleware:** OpenCV, Robot Operating System (ROS), NumPy, scikit-image
- **Embedded Platforms:** Nvidia Jetson (Nano, AGX, NX), Arduino, BeagleBone, DM365, DM8168, IMX6
- **Other Tools:** QGIS, SNAP Toolbox (SAR)
- **Operating Systems:** Linux (embedded & desktop), Windows

REFERENCES

Prof. Koshy George | Co-Principal Investigator, MSIF Lab, Currently at Cambridge Insitute of Technology
kgeorge@ieee.org — Known since 2012 (Master's coursework and Thesis supervisor- CISL PESIT, Co-PI, MSIF Lab, GITAM)

Asst. Prof. Jeevan K. M | Principal Investigator, MSIF Lab, GITAM University
jmani@gitam.edu — Known since 2022